

MODULE 4

BIODIVERSITY & ECOSYSTEM

Conservation of Natural Resources

Course Code: BCV755B

Terra Study Materials
Department of Civil Engineering



Complete Study Notes

Biodiversity • Levels of Biodiversity • Values of Biodiversity • Threats • Conservation (In-situ & Ex-situ) • Ecosystem • Types of Ecosystems • Significance • Economic Values

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1. Biodiversity – Introduction

1.1 What is Biodiversity?

Biodiversity

Biological Diversity, or **Biodiversity**, is that part of nature which includes the differences in genes among individuals of a species, the variety and richness of all plant and animal species at different scales (local, regional, national and global), and the different types of ecosystems — both terrestrial and aquatic — within a defined area. In short, **biodiversity deals with the degree of nature's variety in the biosphere.**

Biodiversity therefore includes three ideas rolled into one:

- The **differences in genes** among the individuals of a species.
- The **variety and richness** of all plant and animal species at different scales in space — locally, in a region, in a country, and in the world.
- The **types of ecosystems**, both terrestrial and aquatic, present within a defined area.

1.2 Flora and Fauna

Term	Meaning
Flora	All the plant life found in a particular region or time period.
Fauna	All the animal life found in a particular region or time period.

Remember

Together, **Flora + Fauna + Microorganisms** form the **biological diversity** of any region.

2. Levels of Biodiversity

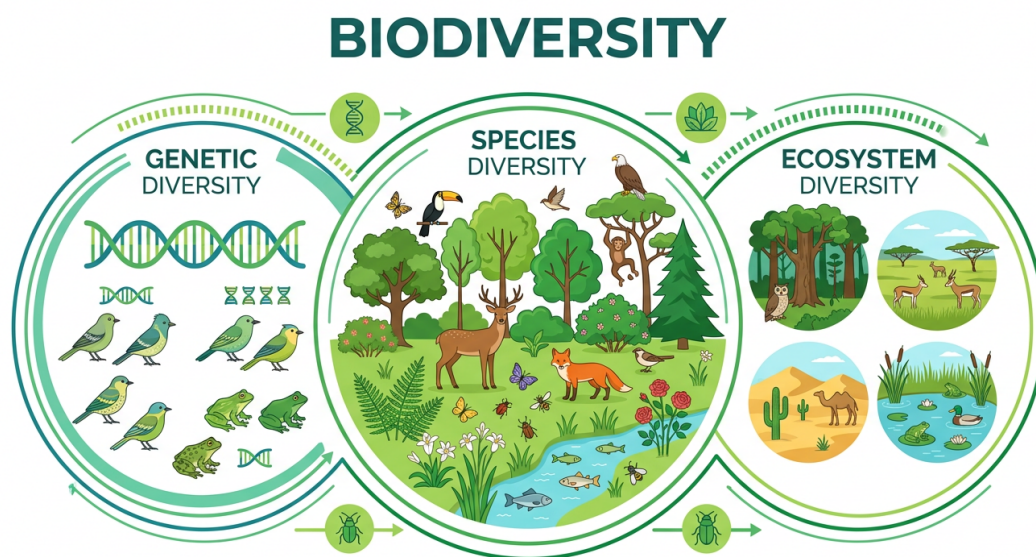
Biodiversity is studied at **three levels**: genetic, species and ecosystem.

2.1 Genetic Diversity

Genetic Diversity

A level of biodiversity that refers to the **total number of genetic characteristics** in the genetic makeup of a species.

- Each member of any animal or plant species **differs widely** from other individuals in its genetic makeup.
- This is due to the **large number of combinations possible in the genes** that give every individual its specific characteristics.
- **Example:** every human being is different from all others.



What biodiversity means – genes, species and ecosystems.

- The diversities present in wild species form the “**gene pool**”.
- Using genetic diversity, more **productive, disease-resistant crops** and superior domestic animals have been developed.
- **Modern biotechnology** also manipulates genes to develop better medicines and industrial products.

Example – One Species, Many Genes

Chihuahuas, Beagles and Rottweilers are all the same species (the dog), yet they look and behave very differently because there is wide **variety in their genes**.

2.2 Species Diversity

Species Diversity

Refers to the **variety of species within a region**. It is an index that incorporates the **number of species** in an area and their **relative abundance**.

- This diversity is seen both in the **natural ecosystem** and in the **agricultural ecosystem**.
- **Example:** monkeys, dragonflies and meadow beauties are all different species.
- Conservation scientists have identified and categorised about **1.8 million species** on Earth.
- Areas rich in species diversity are called “**hotspots of diversity**”.
- **India is among the world’s 15 nations** that are exceptionally rich in species diversity.

2.2.1 Estimated Global Species Numbers

Group	Estimated Number
Total terrestrial species	~8.7 million
Oceanic species	~2.2 million
Vascular plants	~220,000
Marine species	0.7–1 million
Insects	10–30 million
Bacteria	5–10 million
Fungi	1.5–3 million
Mites	~1 million
Fungi documented by 2001	~0.075 million

2.3 Ecosystem Diversity

Ecosystem Diversity

Refers to the **diversity of a place at the level of ecosystems**. It can be described for specific geographical regions, or for political entities such as a country, state or taluk.

- An ecosystem is referred to as “**Natural**” when it is relatively undisturbed by human activities, and “**Modified**” when it is changed to other uses such as farmland or urban areas.
- Life was probably initiated as a product of **organic reactions in Earth’s primordial seas**.
- Alternative possibilities include life beginning in muddy ooze, or life having been seeded from outer space.

3. Value / Importance of Biodiversity

There are **six types of values** of biodiversity:

1. Consumptive use value
2. Productive use value
3. Social value
4. Ethical and moral value
5. Aesthetic value
6. Option value

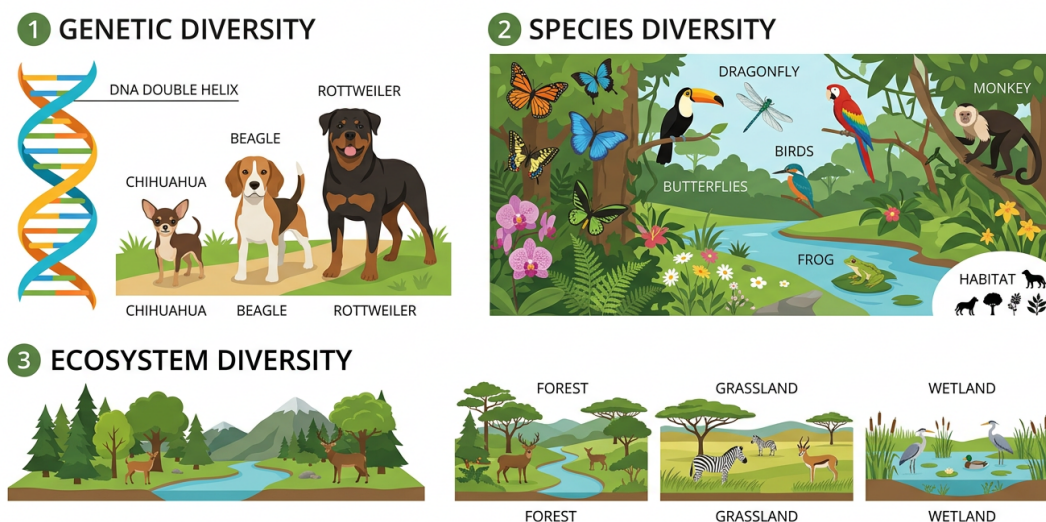
3.1 Consumptive Use Value

- **Direct utilisation** of timber, food, fuel-wood and fodder by local communities.
- Biodiversity provides resources that are **directly consumed** without entering the market.

3.1.1 A. Drugs and Medicinal Plants

- Many plants are used in **primary health care**.

LEVELS OF BIODIVERSITY



The three levels of biodiversity.

- About 70% of modern medicines are derived from plants and plant extracts.

Drug	Plant Source	Use
Atropine	Belladonna	Reduces intestinal pain in diarrhoea
Bromelain	Pineapple	Controls tissue inflammation due to infection
Caffeine	Tea, Coffee	Stimulant of the central nervous system
Camphor	Camphor tree	Increases local blood supply
Cocaine	Cocoa	Analgesic and local anaesthetic
Codeine	Opium poppy	Analgesic – reduces pain
Morphine	Opium poppy	Analgesic – controls pain
Colchicine	Autumn crocus	Anticancer agent
Digitoxin	Common foxglove	Cardiac stimulant used in heart diseases
Diosgenin	Wild yams	Source of female contraceptive
L-Dopa	Velvet bean	Controls Parkinson's Disease
Ergotamine	Smut-of-rye (ergot)	Control of haemorrhage and migraine headaches
Gossypol	Cotton	Male contraceptive
Indicine N-oxide	Heliotropium indicum	Anticancer agent
Penicillin	Fungus	Antibiotic
Quinine	Cinchona bark	Malaria treatment

3.1.2 B. Fuels

- **Fire-woods** are directly consumed by villagers for cooking and heating.

3.1.3 C. Food

- A large number of **wild plants and wild animals** are consumed by human beings as food.

3.2 Productive Use Value

- Biodiversity products have **commercial value** – they are marketed and sold.
- These are derived from both animals and plants.

Animal Products:

- **Silk** from silk worm
- **Wool** from sheep
- **Musk** from musk deer
- **Leather** from animals

Plant Products:

- **Wood** for paper and plywood
- **Cotton** for the textile industry
- **Pearl** for the pearl industry

3.3 Social Value

- Refers to the manner in which **bio-resources are used in society**.
- Associated with the **social life, religion and spiritual aspects** of people.
- **Examples:**
 - **Holy plants:** Tulsi, Lotus, Neem trees
 - **Holy animals:** Cow, sheep, bull, peacock
 - **Holy rivers:** River Ganga
 - **Holy trees:** Tulsi

3.4 Ethical and Moral Value

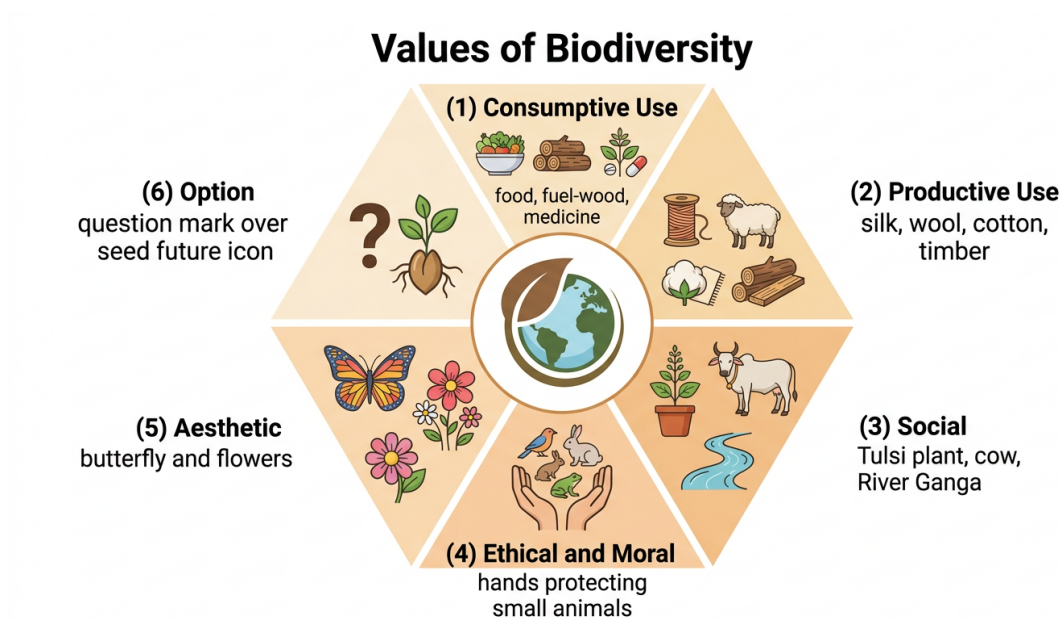
- A species **may or may not be used**, but its existence in nature gives us pleasure.
- Based on the importance of **protecting all forms of life**.
- Most religious and secular beliefs agree that **all forms of life have the right to exist on Earth**.
- Basic philosophy: **“Live and let others Live”**.

3.5 Aesthetic Value

- Aesthetic = appearance / beauty.
- Involves the **appreciation of the presence of biodiversity** for its inherent value and beauty.
- Also for the **contribution it makes to our knowledge, aesthetics, imagination and creativity**.
- The beautiful nature of plants and animals **insists us to protect biodiversity**.
- **Examples:** eco-tourism, the colour of a butterfly, flowers, etc.

3.6 Option Value

- The option value suggests that any species **may be proved to be a valuable species some day**.
- **Keeping future possibilities open** for their use is called the **Option Value**.
- Predicting which present species of crops and domestic animals will be of greatest use in the future is important.



4. Threats to Biodiversity

The Six Main Threats

- | | |
|---|--------------------------|
| 1. Habitat loss (natural calamities and humans) | 4. Exotic species |
| 2. Overpopulation | 5. Overuse of vegetation |
| 3. Environmental pollution | 6. Climate change |

4.1 Habitat Loss

Habitat Loss

Habitat loss can be described as the situation when an **animal loses its home**. Every animal has a **niche** (its role in its community) – without a habitat, there is no niche.

- **Natural causes:** habitat destruction through volcanic activity, fire and climate change.
- **Human causes:** agriculture and farming; harvesting natural resources for personal use; industrial development and urbanisation.

Solutions for Habitat Loss:

- Protecting the **remaining intact sections** of natural habitat.
- Reducing human population and the expansion of urbanisation and industries.
- **Educating the public** about the importance of natural habitat and biodiversity.
- **Planting trees** and home gardens.

4.2 Overpopulation (The Population Connection)

- Past losses of biodiversity can be attributed to the **expansion of the human population** over the globe.
- Continuing growth will **further alter natural ecosystems**, leading to inevitable loss of more wild species and decline in populations.
- One key to holding down biodiversity loss lies in **bringing human population growth down**.

A Frightening Projection

If the human population increases to **10 billion** (as some demographers believe), the consequences for the natural world will be frightening.

4.3 Environmental Pollution

- **Pollution directly kills** many kinds of animals and plants and seriously reduces their populations.
- **Pollution destroys or alters habitats** – with consequences just as severe as those caused by deliberate conversions.
- Most global pollution problems can be traced to the **industrialised world**.

4.4 Climate Change

- Pollution also alters habitats through **climate change**.
- Rising temperatures, changing rainfall patterns and sea-level rise all threaten species and ecosystems.
- It forces species to **migrate or face extinction**.

4.5 Exotic Species (Migrants)**Exotic Species**

An exotic species is a **species introduced into an area from somewhere else** – often from a different continent.

Exotic species threaten biodiversity by:

- **Spreading diseases**
- Acting as **predators or parasites**

- **Competing or hybridising** with native species
- **Altering habitats**

Example – Water Hyacinth

The transportation of species by humans has occurred throughout history. **Water Hyacinth**, introduced into Indian water bodies, now threatens native aquatic life by choking lakes and rivers.

4.6 Overuse of Vegetation

- **Overuse** is another major assault against wild species, responsible for recent extinctions.
- Driven by a combination of **greed, ignorance and desperation**.
- Forms of overuse: **wildlife trade** (hunting and poaching), and the usage of **products derived from wild species**.

THREATS TO BIODIVERSITY



The six threats to biodiversity.

5. Conservation of Biodiversity

5.1 Why Conservation is Needed

- Conservation methods can help **protect and restore ecosystems**.
- It is of utmost importance to develop **quick and effective ways to protect species from extinction**.
- If we fail to conserve species facing extinction now, we are **paving the way for easier and faster depletions in the future**.
- Without conservation, we will live in a **domesticated world lacking wild biodiversity**.

- **WWF (World Wildlife Fund)** and **WCU (World Conservation Union)** are active in conserving biodiversity at the world level.

Forecast of Threatened Species

Estimates forecast a **7% increase** in the number of threatened species by 2020, and a **14% increase** by 2050.

5.2 Two Methods of Conservation

5.2.1 In-Situ Conservation (On-site)

In-Situ Conservation

“Conservation of species **in its natural habitat**, in the place where the species normally occurs.”

- The strategy involves establishing small or large **protected areas**.
- Today in the world there are about **9,800 protected areas** and **1,500 national parks**.

Methods of In-Situ Conservation:

- **(a) Nature or Biosphere Reserves:** large protected areas maintained in their natural state, where both humans and wildlife coexist with minimal interference. *Example: Nilgiri Biosphere Reserve.*
- **(b) National Parks and Wildlife Sanctuaries:** wildlife protected from hunting and habitat destruction. *Examples: Mudumalai, Vedanthangal.*
- **(c) On-Farm and Home Garden Conservation:** conservation of plants, vegetables and fruits; helps maintain traditional crop variety and preserves the genetic diversity of cultivated plants.

5.2.2 Ex-Situ Conservation (Off-site)

Ex-Situ Conservation

Conservation involving the **maintenance and breeding of endangered plant and animal species** under partially or wholly controlled conditions in zoos, gardens and laboratories.

The crucial issue is to **identify those species that are more at risk of extinction**.

Method	Explanation
Long-term Captive Breeding	Animals kept and bred in captivity (zoos) for long-term survival.
Short-term Propagation & Release	Quickly breed species in a lab/zoo and release them into the wild.
Animal Translocation & Re-introduction	Move animals from one location to another; reintroduce them into their natural habitat.
Seed Bank	Seeds stored at low temperature and humidity for long-term preservation of plant genetic material.
Reproductive Technology	Advanced techniques including embryo transfer technology and cloning.

5.3 Other Conservation Methods

5.3.1 National Parks

- Protected areas where **wildlife is conserved**.
- No human habitation, grazing or hunting is allowed; the government manages and protects them.
- **Examples in India:** Jim Corbett, Kaziranga, Sundarbans, Bandipur.

5.3.2 Wildlife Sanctuaries

- Areas where **wild animals are protected** from disturbance and hunting.
- Some human activities may be permitted.
- **Examples:** Mudumalai, Bannerghatta.

5.3.3 Zoological Gardens (Zoos)

- Places where **wild animals are kept in captivity** for conservation, research and education.
- Help in **captive breeding programmes** for endangered species.
- Serve as public education centres about biodiversity.

5.3.4 Gene Banks

- Repositories for **preserving genetic material** (seeds, pollen, sperm, embryos).
- Important for long-term conservation of plant and animal genetic resources.
- **Example:** National Bureau of Plant Genetic Resources (NBPGR), New Delhi.

5.3.5 Pollen Culture

- A technique of **storing pollen grains** at low temperatures for long periods.
- Important for conservation of plant genetic diversity; used in plant breeding programmes.

5.3.6 Ecological Restoration

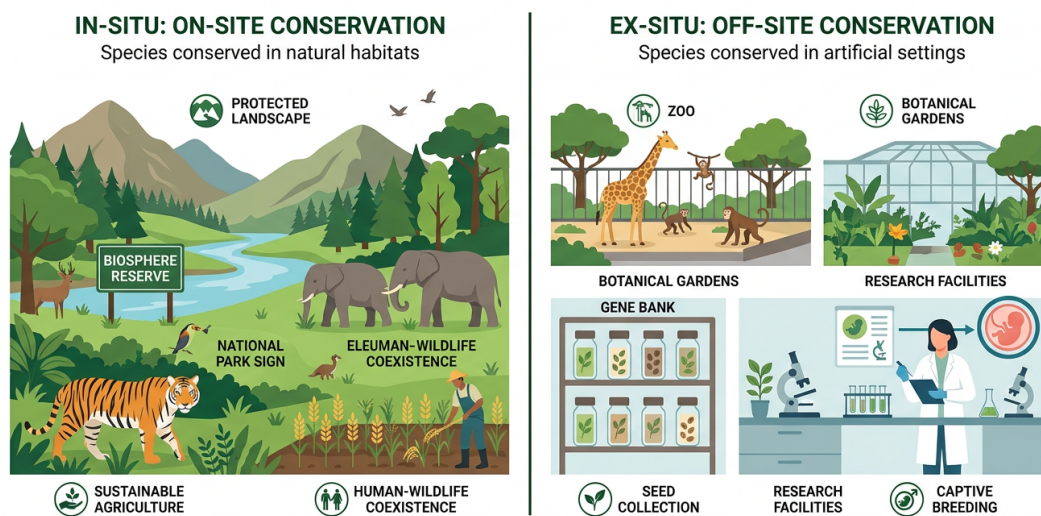
- The process of **assisting the recovery of an ecosystem** that has been degraded, damaged or destroyed.

- Aims to re-establish the structure, function and integrity of the ecosystem.
- Methods include replanting native vegetation, reintroducing native species, and removing invasive species.

5.3.7 Social Forestry

- Management and protection of forests with the **involvement of local communities**.
- Aims to meet local needs for **timber, fuel and fodder** while protecting the environment.
- **Types:** Farm forestry, Extension forestry, Community forestry.
- Helps provide fuel-wood, timber and fodder to villages, reducing pressure on natural forests.

CONSERVATION OF BIODIVERSITY



In-situ versus ex-situ conservation.

6. Ecosystem

6.1 Classification of Ecosystem

Classification Tree

Ecosystem

- **Natural Ecosystem**
 - **Land (Terrestrial):** Forest, Grassland, Desert
 - **Aquatic:** Pond, Ocean, Estuarine
- **Artificial Ecosystem**
 - Cropland Ecosystem

6.2 Definition of Ecosystem

Ecosystem

An ecosystem is a **community of living organisms** (plants, animals, microorganisms) **interacting with each other and with their physical environment** (soil, water, air, temperature, etc.) as a system.

- The term was coined by **Sir Arthur George Tansley** in **1935**.
- Every ecosystem has two main components: **Biotic** (living) and **Abiotic** (non-living).

6.3 Components of Ecosystem

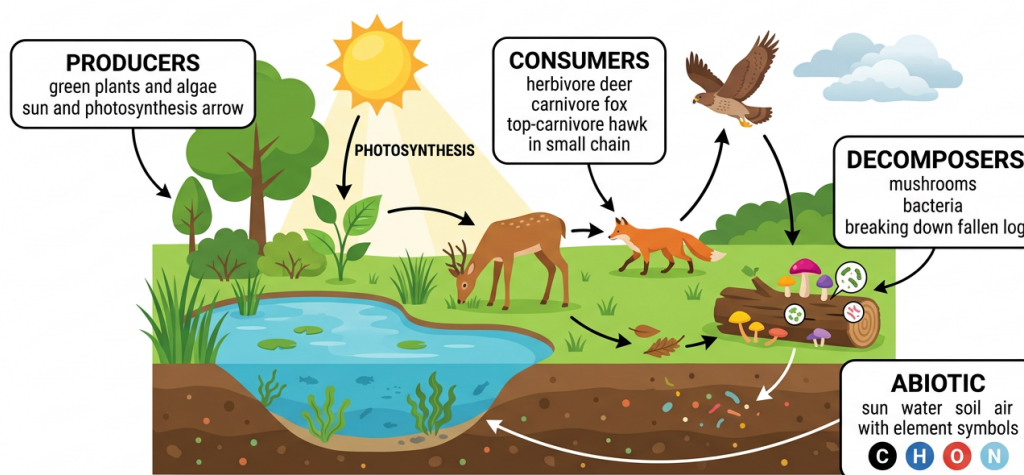
6.3.1 A. Biotic Components (Living)

1. **Producers (Autotrophs)** – plants/algae that make their own food through photosynthesis.
2. **Consumers (Heterotrophs)** – organisms that consume other organisms:
 - Primary consumers (herbivores) – eat plants
 - Secondary consumers (carnivores) – eat herbivores
 - Tertiary consumers (top carnivores) – eat secondary consumers
3. **Decomposers (Saprotrophs)** – break down dead organic matter and recycle nutrients.

6.3.2 B. Abiotic Components (Non-living)

- **Inorganic substances:** C, H, O, N, P, S, Ca, K, Na, Mg.
- **Organic compounds:** proteins, carbohydrates, humus.
- **Physical conditions:** temperature, light, rainfall, humidity.

COMPONENTS OF AN ECOSYSTEM



Components of an ecosystem.

7. Types of Ecosystems

7.1 Forest Ecosystem

Introduction:

- A forest is an area with a **high density of trees**.
- World's total land area: **13,076 million hectares** (FAO, 1989).
- Total forests account for **about 31% of the world's land area**.
- In India, forest cover is **roughly 19%** of the total land area.
- Forest ecosystems are of great concern from the environmental point of view.

Environmental Services:

- Nutrient cycling
- Maintaining biodiversity
- Providing wildlife habitat
- Affecting rainfall patterns
- Regulating stream flow
- Storing water
- Reducing flooding
- Preventing soil erosion
- Reclaiming degraded land

Traditional Values:

- Fire-wood and timber
- Fruits
- Gums
- Herbs and drugs

7.1.1 Structure and Function – Forest Ecosystem

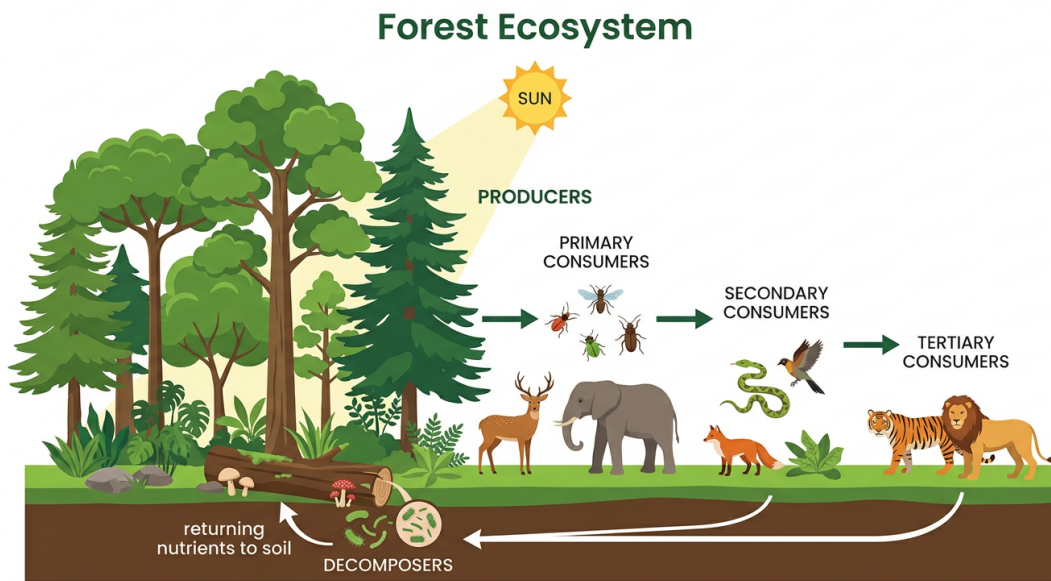
I. Biotic Components

- **Producers:** mainly **trees** (varying with forest type and climate), plus climbers, epiphytes, shrubs and ground vegetation. Dominant species: *Tectona grandis* (Teak), Acer, Betula, Picea, Pine, Cedrus.

Consumer	Category	Examples
Primary	Herbivores feeding on producers	Ants, beetles, bugs, spiders on leaves; elephants, deer, giraffes on shoots and fruits
Secondary	Carnivores feeding on primary consumers	Birds, lizards, frogs, snakes, foxes
Tertiary	Secondary carnivores (top)	Lion, Tiger

- **Decomposers:** saprotrophic microorganisms – Bacteria (*Bacillus*, *Clostridium*, *Pseudomonas*), Fungi (*Aspergillus*, *Ganoderma*, *Fusarium*), Actinomycetes (*Streptomyces*). They attack dead/decayed bodies, releasing nutrients for reuse.

II. Abiotic Components: basic inorganic and organic compounds in soil and atmosphere; dead organic debris littered in the forest.



Structure of a forest ecosystem (energy flow).

7.2 Desert Ecosystem

Introduction:

- A desert is a landscape/region that receives **almost no precipitation** – average annual rainfall **less than 250 mm**.
- Occupies about **17% of Earth's surface**.
- Characterised by **hot days and cold nights** and **scanty flora and fauna**.
- Major deserts: south-western USA and Mexico (North America); Thar, Gobi, Tibet (Asia); West Asia.
- Soils often have **abundant nutrients but little or no organic matter**.

Biotic Components:

- **Producers:** mainly shrubs/bushes, some grasses and a few trees; dominant species are **succulents** (water-retaining plants) and hardy grasses; lower plants include lichens and xerophytic mosses.
- **Consumers:** animals living in **xeric (dry) conditions** – insects, reptiles, nocturnal rodents, birds, camels.
- **Decomposers:** poor, due to low dead organic matter; mostly some bacteria and fungi that are **thermophilic** (heat-loving).

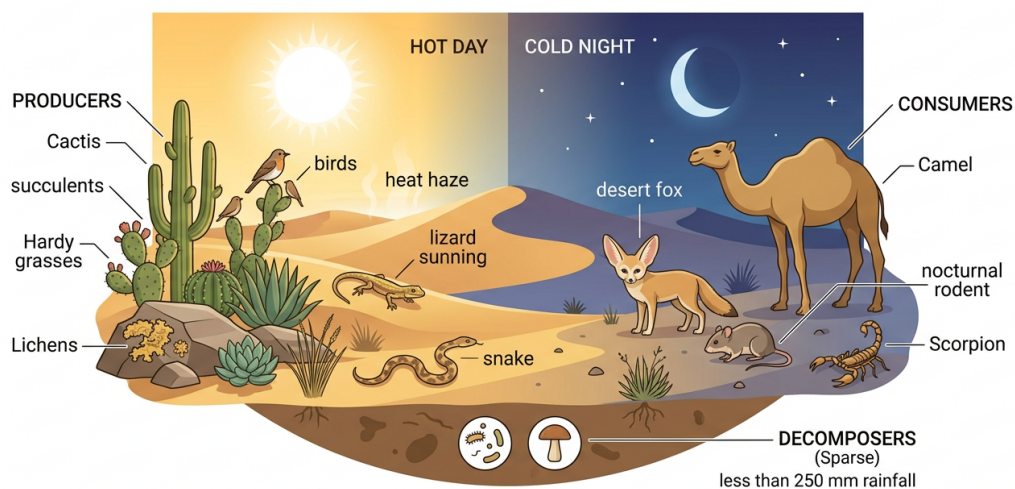
Abiotic Components: due to high temperature and very low rainfall, organic substances are poorly present in the soil.

7.3 Grassland Ecosystem

Introduction:

- Grasslands (also called **Greenswards**) are areas dominated by **grasses and other herbaceous (non-woody) plants**.

Desert Ecosystem



Desert ecosystem.

- Occupy about **24% of Earth's surface**.
- Occur in regions **too dry for forests and too moist for deserts**; annual rainfall **25–75 cm** (usually seasonal).
- Show the **highest abundance and greatest diversity of large mammals**.

Grassland	Location
Prairies	Canada, USA
Pampas	South America
Steppes	Europe and Asia
Veldts	Africa

- Dominant animals: wild horses, asses and antelope of Eurasia; herds of Bison of America.

Biotic Components:

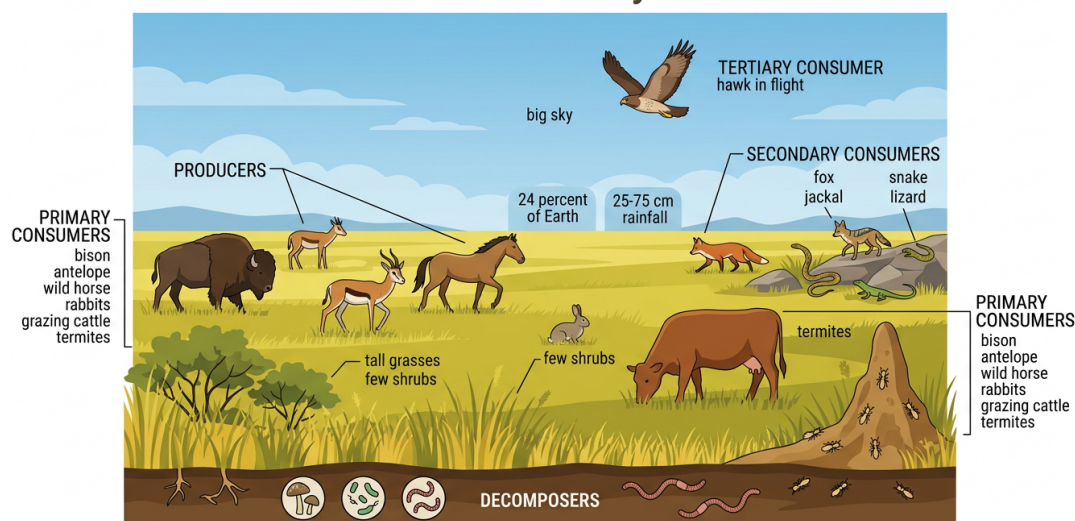
- **Producers:** mainly grasses, plus a few herbs and shrubs – e.g. *Brachiaria*, *Cynodon*, *Desmodium*, *Digitaria*.

Consumer	Category	Examples
Primary	Herbivores feeding on grasses	Cows, buffaloes, sheep, goats, deer, rabbits, insects, termites
Secondary	Carnivores feeding on herbivores	Frogs, snakes, lizards, birds, foxes, jackals
Tertiary	Feed on secondary consumers	Hawks

- **Decomposers:** saprotrophic Bacteria, Fungi and Actinomycetes that decompose dead matter and release nutrients for reuse.

Abiotic Components: inorganic and organic compounds in soil and air; elements such as C, H, N, O, P, S supplied by water, nitrates, sulphates and phosphates.

Grassland Ecosystem



Grassland ecosystem.

7.4 Aquatic Ecosystems

Introduction:

- Aquatic ecosystems deal with the **biotic community present in water bodies**.
- In terrestrial ecosystems CO_2 and O_2 are in **gaseous form**; in aquatic ecosystems they are available in the **dissolved state**.

Classification: (1) Freshwater Ecosystem, (2) Marine Ecosystem.

7.4.1 Freshwater Ecosystem

- Cover **0.8% of Earth's surface** and contain **0.009% of its total water**.
- Contain **41% of the world's known fish species**.
- Functions: recycle nutrients, purify water, attenuate floods, recharge groundwater, and provide habitats for wildlife; important for human recreation and tourism.

Type	Description
Lentic	Slow-moving water – pools, ponds and lakes.
Lotic	Rapidly-moving water – streams and rivers.
Wetlands	Areas where the soil is saturated with water or inundated for at least part of the time.

7.4.2 Lakes and Pond Ecosystem (Lentic)

- A pond is a place where living organisms **not only live but interact** with biotic and abiotic components.

- Ponds are often under **tremendous anthropogenic pressure**.
- Lakes are usually **big standing freshwater bodies**.

Zone	Description
Littoral Zone	Shallow water zone near the shore.
Limnetic Zone	Open water zone with effective penetration of solar light (high productivity); also called the Euphotic Zone.
Profundal / Aphotic Zone	Deep water zone where light penetration is low (little productivity).

Biotic Components:

- **Producers:** submerged, free-floating and amphibious macrophytes (Hydrilla, Utricularia, Wolfia, Azolla, Typha); minute phytoplankton (Ulothrix, Spirogyra, Oedogonium).
- **Consumers:** primary – zooplankton (ciliates, flagellates, protozoa, small crustaceans) and benthos; secondary – insects and fishes; tertiary – large fishes feeding on small fishes.
- **Decomposers:** bacteria, fungi and actinomycetes.

Abiotic Components: inorganic and organic substances in bottom soil or dissolved in water, plus some dead organic matter.

7.4.3 Marine / Ocean Ecosystem

- Includes oceans, estuaries and lagoons, mangroves, coral reefs, deep sea and sea floor.
- Covers approximately **71% of Earth's surface** (about **361 million sq km**).
- Differs from freshwater mainly because of its **salty water – 3.5% salt (35 ppt)**; dominant ions **Sodium and Chloride**.
- Average temperature **2–3°C**; deep areas are devoid of light.

Biotic Components:

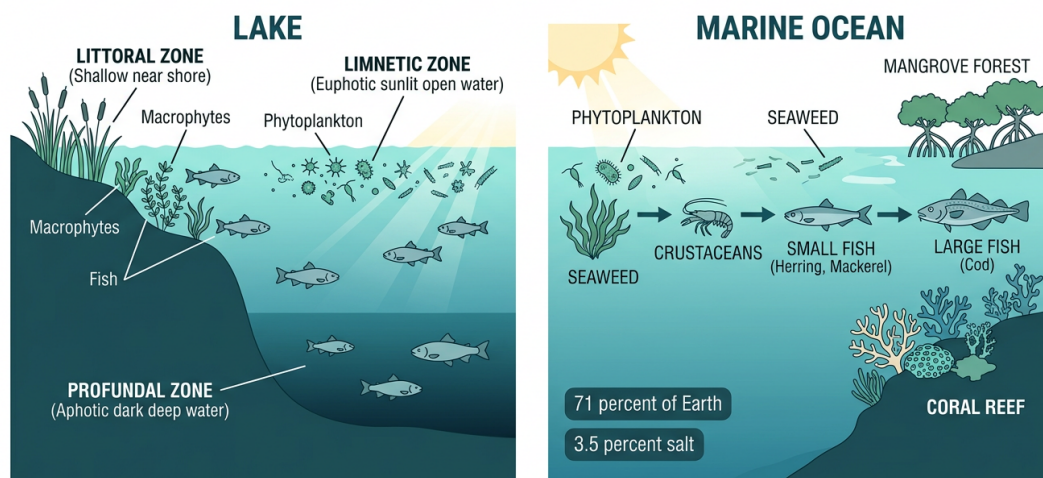
- **Producers:** phytoplankton (Diatoms, Dinoflagellates), large seaweeds (Chlorophyceae, Phaeophyceae, Rhodophyceae), angiosperms (Ruppia, Zostera, Posidonia) and mangrove vegetation (Rhizophora, Carapa).
- **Consumers:** primary – crustaceans, mollusks, fish; secondary – carnivorous fishes (Herring, Shad, Mackerel); tertiary – top carnivorous fishes (Cod, Haddock).
- **Decomposers:** bacteria and fungi.

Abiotic Components: high concentrations of Na, Ca, Mg and K salts; variable dissolved oxygen; light and temperature create unique physio-chemical conditions.

7.5 Estuarine Ecosystem

- An **estuary** is a coastal area where **freshwater from rivers meets saltwater from the sea**, creating a transition zone.
- Characterised by **variable salinity** (brackish water).
- Highly productive – serve as **nurseries for many marine species**.

AQUATIC ECOSYSTEMS



Aquatic ecosystems – freshwater lake zones and the marine ocean.

- Support rich biodiversity: mangroves, fish, birds, crustaceans.
- **Examples in India:** Sundarbans Delta, Chilika Lake.

7.6 Wetland Ecosystem

- Areas where the **soil is saturated with water or inundated** for at least part of the time; very biologically diverse.
- **Functions:** water purification, flood control, carbon sequestration, and habitat for birds, fish and amphibians.
- **Examples:** Chilika Lake (Odisha), Loktak Lake (Manipur), Keoladeo National Park (Rajasthan).

8. Significance of Ecosystems

8.1 Why Ecosystems are Important

1. **Food relationships** among organisms form the basis of all ecological understanding.
2. **Food chains are the living components of the biosphere** – they represent the flow of energy and matter.
3. **Food chains are vehicles of transfer of energy** from one trophic level to the next:

Producers → Primary → Secondary → Tertiary Consumers

At each level, approximately **10% of energy is transferred** (about 90% is lost as heat).

4. Through food chains, **transfer of materials and nutrients** also takes place – this is called **biogeochemical cycling**.
5. The movement of **toxic substances** (like **DDT**) can be studied through food chains:

- DDT sprayed to kill pests enters the food chain.
- It accumulates at higher trophic levels through a process called **biomagnification**.
- At the highest trophic level the concentration becomes **dangerously high**.

8.2 Biogeochemical Cycling

Biogeochemical Cycle

The **cycling of chemical elements** through the biotic (living) and abiotic (non-living) components of an ecosystem.

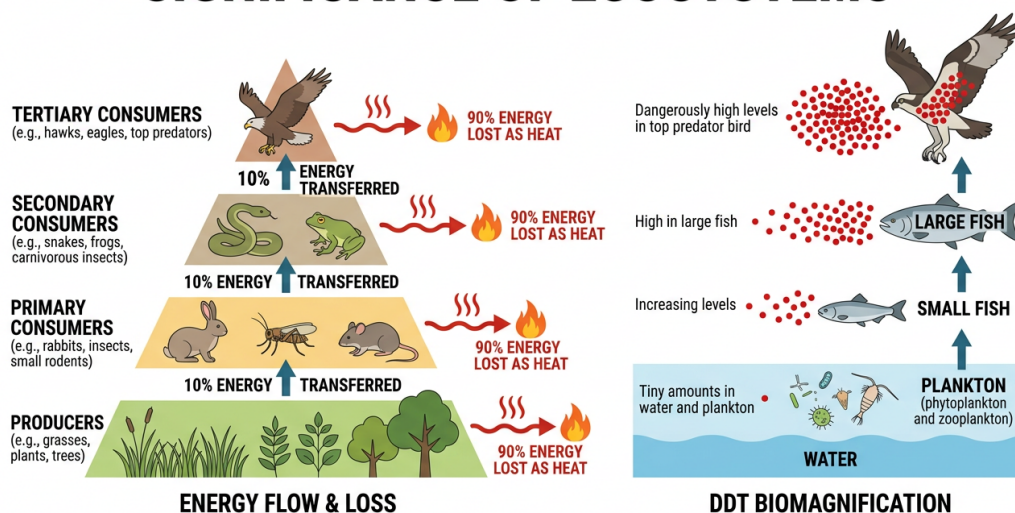
Biodiversity plays a crucial role in:

- **Nitrogen cycle** – nitrogen-fixing bacteria, decomposers.
- **Carbon cycle** – photosynthesis, respiration, decomposition.
- **Phosphorus cycle** – plants, decomposers, soil.
- **Water cycle** – transpiration by plants, evaporation.

Why it Matters

Without biodiversity, these cycles would be disrupted, leading to the collapse of life-supporting systems.

SIGNIFICANCE OF ECOSYSTEMS



Energy flow and biomagnification in a food chain.

9. Economic Values – Medicinal Plants, Drugs and Fisheries

9.1 Medicinal Plants

- About **70% of modern medicines** are derived from plants.

- Traditional systems like **Ayurveda, Unani and Siddha** rely heavily on plant-based medicines.
- Key examples:
 - **Penicillin** (fungus) – antibiotic
 - **Quinine** (Cinchona bark) – malaria treatment
 - **Morphine** (opium poppy) – analgesic
 - **Aspirin** (from willow bark) – pain reliever
 - **Taxol** (from Yew tree) – anticancer drug

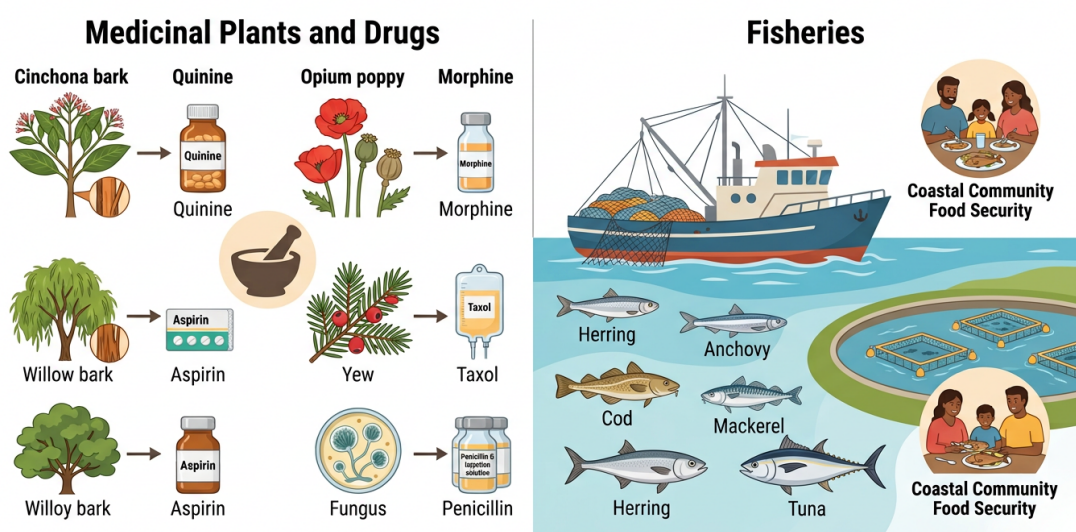
9.2 Drugs from Biodiversity

A detailed list of commonly used modern drugs derived from plant sources is given in the table under **Consumptive Use Value** (Section 3).

9.3 Fisheries and Biodiversity

- **Fisheries** represent a major economic sector supported by biodiversity.
- Freshwater ecosystems contain **41% of the world's known fish species**.
- Marine ecosystems support a massive **commercial fishing industry** – major fish include Herring, Mackerel, Cod, Haddock and Tuna.
- **Biodiversity ensures healthy fish populations** – the loss of any species disrupts the food web.
- Fisheries provide **food security** for millions, especially in coastal communities.
- **Aquaculture (fish farming)** also relies on biodiversity.

Economic Values of Biodiversity



Economic value of biodiversity – medicines and fisheries.

9. Summary Table – Module 4: Biodiversity & Ecosystem

Topic	Key Concept
Biodiversity	Variety and richness of all plant and animal species and ecosystems; deals with nature's variety in the biosphere.
Flora and Fauna	Flora = plant life; Fauna = animal life of a region.
Levels of Biodiversity	Genetic (gene differences), Species (variety of species), Ecosystem (diversity of ecosystems).
Genetic Diversity	Differences in genes among individuals; forms the 'gene pool'; basis for biotechnology.
Species Diversity	Number + relative abundance of species; ~1.8 million identified; India is a biodiversity hotspot.
Ecosystem Diversity	Diversity of ecosystems – natural vs modified.
Value of Biodiversity	Consumptive (food, fuel, drugs), Productive, Social, Ethical, Aesthetic, Option.
Medicinal Plants	70% of modern medicines from plants – Penicillin, Quinine, Morphine, etc.
Threats to Biodiversity	Habitat loss, overpopulation, pollution, climate change, exotic species, overuse.
In-Situ Conservation	In natural habitat – Biosphere reserves, National parks, Sanctuaries, On-farm conservation.
Ex-Situ Conservation	Off-site – Captive breeding, Seed banks, Zoological gardens, Reproductive technology.
National Parks	Fully protected areas – Jim Corbett, Kaziranga, Bandipur.
Wildlife Sanctuaries	Protected areas with limited human activity – Mudumalai, Bannerghatta.
Gene Banks	Storage of genetic material – seeds, pollen, embryos.
Social Forestry	Community-managed forestry for fuel, fodder and environmental protection.
Ecosystem Definition	Community of organisms interacting with each other and their environment (Tansley, 1935).
Forest Ecosystem	31% of world's land; biotic: trees, herbivores, carnivores, decomposers.
Desert Ecosystem	<250 mm rainfall; 17% of Earth; succulents, reptiles, camels.
Grassland Ecosystem	24% of Earth; 25–75 cm rainfall; grasses, herbivores, carnivores.

Topic	Key Concept
Freshwater Ecosystem	Lentic (ponds/lakes), Lotic (rivers/streams), Wetlands; 41% of fish species.
Marine Ecosystem	71% of Earth's surface; 3.5% salt; phytoplankton, seaweeds, fish.
Estuarine Ecosystem	Where freshwater meets saltwater; highly productive; mangroves.
Biogeochemical Cycling	Cycling of nutrients (N, C, P, water) through biotic and abiotic components.
Significance of Ecosystems	Energy transfer through food chains; nutrient cycling; study of biomagnification.

Module Complete

These are complete, detailed study notes for **Module 4: Biodiversity and Ecosystem** (BCV755B – Conservation of Natural Resources), covering every topic of the module without skipping anything.